# Measuring the radius of a star — Solution

Y23 (2023) — English translation

**A1** **0.20 pts**

*Let us consider the case when rays 1 and 2 are parallel to the $z$ axis, and mirror M3 is parallel to the $xy$ plane. Find the difference in optical paths $\Delta_{12}$ between rays 1 and 2 if they both fall on the screen at point $O$.*

In horizontal sections, the path difference is $L_1 - L_2$, and the first ray also travels a distance $L_3$ twice. Thus, the total difference is:

$$\Delta_{12} = L_1 + 2L_3 - L_2$$

**Answer:** \textbfAnswer:

$$\Delta_{12} = L_1 + 2L_3 - L_2$$

**A2** **1.40 pts**

*Let us consider the case when the mirror M3 is deflected by an angle $\alpha$ from the plane $xy$ by rotating around the axis $x$ (see Fig. 1). If ray 1' hits the screen at point $O$, find the optical path difference $\Delta_{1'2}$ between rays 1' and 2. Show that, to a first approximation, the path difference is independent of $\alpha$ and simplify your answer at $\alpha \ll 1$.*

Note that the triangle obtained from the paths of rays 1 and 1' from the mirror M3 to the point $O$ is isosceles. Also, changes in the horizontal and vertical paths due to changes in the point of incidence of the beam compensate each other. The only parameter affecting the path traveled by beam 1 is the coordinate $z$ of the point of impact on the mirror M3: it affects the distance traveled from HM to M3. But the optical path of beam 2 does not change in any way. We get:

$$\Delta_{1'2} = \Delta_{12} - (2(L_3 + L_4)\sin\alpha)\cdot\sin\alpha = \Delta_{12} - 2(L_3 + L_4)\sin^2\alpha$$

Substituting result прошлого parta:

$$\Delta_{12'} = L_1 + 2L_3 - L_2 - 2(L_3 + L_4)\sin^2\alpha \approx L_1 + 2L_3 - L_2 - 2(L_3 + L_4)\alpha^2$$

**Answer:** \textbfAnswer:

$$\Delta_{1'2} = L_1 + 2L_3 - L_2 - (L_3 + L_4)\frac{\sin\alpha\sin 2\alpha}{\cos\alpha} = L_1 + 2L_3 - L_2 - 2(L_3 + L_4)\alpha^2$$

**B1** **0.80 pts**

*Let rays 1 and 2 fall at fixed points of mirrors M1 and M2, respectively. Show in the solution sheets that, to a first approximation, the optical paths of rays 1 and 2 from the points of reflection from the mirrors M1 and M2, respectively, to the screen do not depend on the angles $\theta$ and $\alpha$.*

## B2 — 0.80 pts

*Let the angles $\alpha$ and $\theta$ be fixed, and the positions of the points of incidence of rays 1 and 2 on the mirrors M1 and M2, respectively, can change. Show in the solution sheets that the optical paths of rays 1 and 2 from the dotted line corresponding to the coordinate $z_n = L_3 + L_4$ (meaning the points of intersection of the dotted line with the trajectories of rays 1 and 2 before falling on the mirrors M1 and M2, respectively) to the screen, to a first approximation, do not depend on the angles $\theta$ and $\alpha$.*

## B3 — 1.00 pts

*Express $b$ and $\delta_O$ in terms of $\lambda$, $L_1$, $L_2$, $L_3$, $L_4$.*

$\delta_O$ can be found using the previous part of the problem, since this is the value of the phase difference for $\theta = 0$. Note that the phase difference arises not only due to the difference in optical paths, but also due to reflections: beam 1 is reflected one time more, so it is additionally shifted by $\pi$.

$$\delta_O = \frac{2\pi}{\lambda}(L_1 + 2L_3 - L_2) + \pi$$

Можно заметить, что в силу утверждений, доказанных в B1 and B2 for/at изменении угла $\theta$ сность фаз возникает только from-за изменения путей до пересечения с partandрной линией $z_п = L_3 + L_4$, that is:

$$\frac{2\pi}{\lambda}(L_1 + L_2)\theta = \frac{2\pi}{\lambda}b\theta$$

$$b = L_1 + L_2$$

**Answer:** \textbfAnswer:

$$b = L_1 + L_2 \delta_O = \frac{2\pi}{\lambda}\left[L_1 + 2L_3 - L_2\right] + \pi$$

## B4 — 0.60 pts

*Show in the solution sheets that*

$$\delta_y = \delta_O - \frac{2\pi}{\lambda} \cdot 2\alpha y$$

*Note: Further, in all points of the problem, you can use the statements of points B1-B4, even if they are not proven.*

Note: Further, in all points of the problem, you can use the statements of points B1-B4, even if they are not proven.

Consider two rays 1 that arrive at points on the screen located at a distance $y$. Due to the smallness of $\alpha$, the distance between the points of reflection of these rays from M3 is also equal to $y$. Note that the difference in optical paths between them arises only because of the different coordinates $z$ of the reflection points from M3: the path from HM to M3 and from M3 to the screen is shortened.

$$\Delta L = -2y\alpha$$

$$\delta_y = \delta_O - \frac{2\pi}{\lambda} \cdot 2\alpha y$$

**Answer:** \textbfAnswer:

$$\delta_y = \frac{2\pi}{\lambda}\left[L_1 + 2L_3 - L_2 - 2\alpha y\right] + \pi$$

## B5 0.60 pts

*Thus, an interference pattern will appear on the screen. Find the distance $y_1$ to the interference maximum closest to point $O$. Also find the distance $\Delta y_m$ between two adjacent interference maxima. Give numerical answers in the case of $L_1 = 0.50$ m, $L_2 = 1.50$ m, $L_3 = 0.50$ m, $L_4 = 1.00$ m, $\alpha = 10^{-4}$, $\theta = 10^{-5}$, $\lambda = 500$ nm.*

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Substituting numerical values, we get:

$$\delta = 81\pi - 800\frac{\pi}{1\text{m}} \cdot y$$

Максимум соответсвует $\delta = 2\pi n$. At the point $O$ находится минимум.

$$y_1 = \frac{\lambda}{4\alpha} = 1.25\text{mm}$$

$$\Delta y_m = \frac{\lambda}{2\alpha} = 2.5\text{mm}$$

**Answer:** \textbfAnswer:

$$y_1 = 1.25 \cdot 10^{-3}\ \text{m}\Delta y_m = \frac{\lambda}{2\alpha} = 2.5\ \text{mm}$$

## B6 0.90 pts

*Find how the intensity $I(y)$ on the screen depends on the coordinate $y$ if its maximum value is $I_0$, substituting here the numerical values from the previous paragraph. Consider the effect of the beam splitter on the intensity of transmitted light.*

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The field amplitude in beam 1 is $\sqrt{2}$ times less than in the second. Then the intensity is proportional to the square of the amplitude:

$$I \sim 1 + 2 + 2\sqrt{2}\cos\delta$$

$$I(y) = I_0 \frac{3 - 2\sqrt{2}\cos\left[80\pi\left(1 - 10\frac{y}{1\text{m}}\right)\right]}{3 + 2\sqrt{2}}$$

**Answer:** \textbfAnswer:

$$I(y) = I_0 \frac{3 - 2\sqrt{2}\cos\left[80\pi\left(1 - 10\frac{y}{1\text{m}}\right)\right]}{3 + 2\sqrt{2}}$$

## C1 0.40 pts

*Express the wave vector $\vec{k}$ of the incident rays in terms of $\theta_x$, $\theta_y$, the modulus of the wave vector $k = 2\pi/\lambda$ and the unit unit vectors of the coordinate axes.*

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Due to the smallness of the angles $\theta_x$ and $\theta_y$:

$$k_z = -k$$

$$k_x = -k\theta_x$$

$$k_y = -k\theta_y$$

**Answer:** \textbfAnswer:

$$\vec{k} = -k\left(\theta_x\hat{x} + \theta_y\hat{y} + \hat{z}\right)$$

## C2 0.60 pts

*Express the phase difference between beams 1" and 2" in terms of $\lambda$, $L_1$, $L_2$, $L_3$, $L_4$, $\theta_x$, $\theta_y$, $\alpha$, $y$.*

Note that the initial displacement along the $x$ axis of the rays hitting one point on the screen is of the order of $L_3\theta_x$, the difference in the optical paths is correspondingly proportional to $\theta_x^2$, so the contribution $\theta_x$ can be ignored. The answer is similar to the previous part of the problem:

$$\delta_{1''2''} = \frac{2\pi}{\lambda}\big((L_1+L_2)\theta_y + (L_1+2L_3-L_2) - 2\alpha y\big) + \pi$$

**Answer:** \textbfAnswer:

$$\delta_{1''2''} = \frac{2\pi}{\lambda}\big((L_1+L_2)\theta_y + (L_1+2L_3-L_2) - 2\alpha y\big) + \pi$$

## D1 1.80 pts

*Write $V$ as an integral over one variable. It is not necessary to perform integration explicitly. Hint: the integral of some function $f(\theta_y)$, depending only on $\theta_y$, over the surface of a disk with radius $\theta_R$ can be represented as:*

$$I = \int\limits_{-\theta_R}^{\theta_R} \mathrm{d}\theta_y \int\limits_{-\sqrt{\theta_R^2-\theta_y^2}}^{\sqrt{\theta_R^2-\theta_y^2}} \mathrm{d}\theta_x\ f(\theta_y) = \int\limits_{-\theta_R}^{\theta_R} \mathrm{d}\theta_y \cdot 2\sqrt{\theta_R^2-\theta_y^2} f(\theta_y).$$

Hint: the integral of some function $f(\theta_y)$, depending only on $\theta_y$, over the surface of a disk with radius $\theta_R$ can be represented as:

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Parts of the star emit independently, so the intensities from them can be added.

$$dI = Jd\theta_x d\theta_y \left(3 - 2\sqrt{2}\cos\left(\frac{2\pi}{\lambda}\big(L_1+2L_3-L_2-2\alpha y\big) + \frac{2\pi}{\lambda}(L_1+L_2+d)\theta_y\right)\right)$$

Denote $\varphi_0 = \frac{2\pi}{\lambda}\big(L_1+2L_3-L_2\big)$.

$$I = \int\limits_{-\theta_R}^{\theta_R} J\sqrt{\theta_R^2-\theta_y^2}\left(3 - 2\sqrt{2}\cos\left(-\frac{4\pi}{\lambda}\alpha y + \frac{2\pi}{\lambda}(L_1+L_2+d)\theta_y + \varphi_0\right)\right) d\theta_y$$

Note that:

$$I_{max} = \int\limits_{-\theta_R}^{\theta_R} J\sqrt{\theta_R^2-\theta_y^2}\left(3 + 2\sqrt{2}\cos\left(\frac{2\pi}{\lambda}(L_1+L_2+d)\theta_y\right)\right) d\theta_y$$

$$I_{min} = \int\limits_{-\theta_R}^{\theta_R} J\sqrt{\theta_R^2-\theta_y^2}\left(3 - 2\sqrt{2}\cos\left(\frac{2\pi}{\lambda}(L_1+L_2+d)\theta_y\right)\right) d\theta_y$$

Then видность:

$$V = \frac{2\sqrt{2}\int\limits_{-\theta_R}^{\theta_R}\sqrt{\theta_R^2-\theta_y^2}\cos\left(\frac{2\pi}{\lambda}(L_1+L_2+d)\theta_y\right)d\theta_y}{3\int\limits_{-\theta_R}^{\theta_R}\sqrt{\theta_R^2-\theta_y^2}d\theta_y}$$

Denote $\omega$ =Note that the integrable function is even, therefore:

$$V = \frac{8\sqrt{2}}{3\pi} \int\limits_0^1 \mathrm{d}w \sqrt{1-w^2} \cos(\eta w)$$

**Answer:** \textbfAnswer:

$$V = \frac{8\sqrt{2}}{3\pi} \int\limits_0^1 \mathrm{d}w \sqrt{1-w^2} \cos(\eta w)$$

## D2 — 0.90 pts

*Find Betelgeuse's angular radius $\theta_R$. Find its radius $R$ if the distance to it is $h = 6.079 \cdot 10^{15}$ km. Hint: The first two zeros of the integral*

$$F_n = \int_0^1 \left(1-w^2\right)^n \cos(\eta w)\ \mathrm{d}w$$

*as a function of $\eta$ are given in the table below.*

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Hint: The first two zeros of the integral

$$F_n = \int_0^1 \left(1-w^2\right)^n \cos(\eta w)\ \mathrm{d}w$$

as a function of $\eta$ are given in the table below.

The first visibility resets correspond to $\eta_1$ and $\eta_2$ for $n = 0.5$. Thus:

$$\eta_2 - \eta_1 = \frac{2\pi}{\lambda}(d_2 - d_1)\theta_R$$

$$\theta_R = \frac{(\eta_2 - \eta_1)\lambda}{2\pi(d_2 - d_1)} = 1.18 \cdot 10^{-7}$$

$$R = \theta_R h = 7.16 \cdot 10^8 \text{km}$$

**Answer:** \textbfAnswer:

$$\theta_R = \frac{(\eta_2 - \eta_1)\lambda}{2\pi(d_2 - d_1)} = 1.18 \cdot 10^{-7}$$

$$R = \theta_R h = 7.16 \cdot 10^8 \text{km}$$